

Article

Active Equalization of Lithium-Ion Battery Based on Reconfigurable Topology

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Abstract: The equalization technique is a key technique in the secondary utilization of retired batteries. In this paper, a double-layer equalization method is proposed, which combines the reconfigurable topology with the converter active equalization method. The inner layer uses the reconfigurable topology to have a balanced set of battery cells. Thanks to isolating the lowest SOC (state of charge) cell in the battery group, the energy transfer loss among cells is avoided. In addition, this topology can reduce cost and control complexity and the number of components. In the outer layer, a Buck–Boost converter is added for each battery group, and the outputs of the converters are connected in series. The output voltage of the converter varies as the SOC of the group varies while the total output voltage is stable. In order to validate the proposed method, an equalization circuit consisting of 12 battery cells is built on Matlab/Simulink. Simulation results show that the proposed method can effectively balance the battery pack and maintain a stable output voltage. Compared to the conventional active equalization method, the proposed method has significantly improved the equalization efficiency.

Keywords: active equalization; reconfigurable topology; double-layer model; retired battery; secondary utilization; buck–boost



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1. Introduction

The number of electric vehicles (EVs) has increased rapidly to reduce atmospheric pollution in recent years. However, power batteries on EVs are suggested to be replaced when their capacities decline to less than 80% of the initial capacity to ensure the range of EVs. The number of retired power batteries in China exceeded 200,000 tons (about 26 GWh) in 2020 and is expected to rise to about 780,000 tons (about 134 GWh) in 2025, according to data from China Automotive Technology and Research Center. It will cause much waste and heavily harm the environment if these retired batteries are scrapped directly [1]. Retired batteries can be used in energy storage systems in power grids or microgrids when their capacities are greater than 60% of the initial states [2]. The energy storage systems can not only provide energy transmission between interconnected power grids but also provide continuous power to local power grids when faults occur, thus improving the robustness of the power system when confronting extreme weather or sudden faults [3].

Lithium-ion batteries are usually connected in series and parallel to meet the voltage and power requirements of loads because of the low voltage and capacity of one single battery [4]. The origination of inconsistency of lithium-ion batteries can usually be separated into the production process and the usage process [5]. The former can be caused by variances in the raw materials and manufacturing equipment and procedures, which will lead to variances in physical volume, internal impedance, and self-discharge rate [6]. The self-discharge of lithium-ion batteries means that the charge will gradually decrease even if no load is connected, which will accumulate a lot over a long time and cause significant imbalances in SOC [7,8]. The latter mainly originate from environmental differences during battery usage and storage. The recoverable power and capacity can be reduced significantly

when these batteries are operated or stored at temperatures above 50 °C due to multiple factors, including loss of lithium from repair of solid electrolyte interphase (SEI) on negative electrode [9]. Charging at low temperatures may result in lithium deposition and even the growth of dendrites [10]. Additionally, the history of working in high temperatures would aggregate the battery's self-discharge rate permanently [11]. The battery with the lowest capacity determines the amount of electricity that can be discharged from the series battery group. The lowest-capacity battery may be overcharged during charging, which may lead to overheating and even cause an explosion [12]. These problems are more obvious in the secondary utilization of retired batteries since the inconsistency among cells is aggravated by the scattered sources, different production conditions, performance degradation on vehicles, etc. The equalization technique is essential to eliminate the influence of more discrete voltage, internal resistance, and capacity to ensure the available capacity and safety of the battery pack.

The equalization methods of lithium-ion batteries can be divided into active methods and passive methods. Passive methods use resistors connected in parallel with the batteries to dissipate excess electricity to balance the battery pack [13]. This kind of method has a simple structure and low cost, but it wastes energy and raises the temperature in high-power situations. Active methods mainly use capacitors, inductors, transformers, and converters as energy transfer devices [14,15]. In the simplest active equalization circuit, the balancing current can only be transmitted between adjacent battery cells. In that case, if the battery cell with a high imbalance appears at one end of the battery pack, the energy to be equalized will be transmitted over a longer distance in the pack, which will affect the equalization speed and efficiency [16]. Ref. [17] proposed a method combining Buck–Boost converters with switching matrices, which reduces the number of components and cost compared with conventional inductor equalization circuits, but the balancing speed is relatively slow. Ref. [18] put forward a DSSCE (Delta-Structured Switched-Capacitor Equalizer) circuit, which sets a separate capacitor balancing circuit for every two adjacent cells. Although the equalization speed and efficiency are improved, the complexity and cost of the system are greatly increased. The method proposed in Ref. [19] uses forward–flyback converter as the energy transfer device and can automatically balance the cells at any position in the group during charging or discharging, but the cost is high.

In conventional equalization circuits, the energy to be equalized transmits from cell to cell, which leads to extra energy loss. The failure of one battery cell will lead to the failure of the whole battery pack because the cells in the battery pack are permanently connected in series [20]. Therefore, the conventional active equalization methods are not suitable for retired battery energy storage systems, which are highly unbalanced and require high robustness. To solve these problems, in this study, we adopted a new battery balancing topology called reconfigurable topology, which can isolate the lowest SOC (state of charge) battery cell or the faulty battery cell without affecting the normal operation of other cells in the pack.

The earliest reconfigurable equalization topology provides the flexibility to connect or isolate any single cell from the pack while allowing arbitrary cell connections in series, parallel, or hybrid configurations. Although this topology has high flexibility, its cost, energy loss, and complexity are relatively high because every single cell needs five MOSFETs to control [21]. Later, new reconfigurable equalization topologies were developed, including series topology, DESA (Dependable, efficient, scalable architecture) topology, graph-based topology, and matrix topology [22–24]. Each topology has its advantages and disadvantages in terms of cost, flexibility, speed, efficiency, and control complexity.

This paper develops the reconfigurable topology into a double-layer equalization method to enhance the efficiency and robustness of the equalization process of retired batteries. A series reconfigurable equalization topology with two MOSFETs per battery cell is selected as the topology of the inner layer. It is suitable for large-scale energy storage applications because it can improve the balancing efficiency with low cost and control complexity. Buck–Boost converters are chosen in the outer layer to solve the problem

that the single reconfigurable circuit cannot output a stable voltage. The improvement of the equalization efficiency by the proposed method is verified by calculation in Section 4. Finally, the feasibility of the proposed method is validated by the simulation of a 4-series 3-parallel equalization circuit on MATLAB/Simulink in Section 5.

In this paper, the battery pack means the whole battery system, a battery group is a group of n battery cells, and the working part of a battery group consists n or $n - 1$ battery cells depending on the phase of equalization. Here are some notations used in the rest of this paper.

- (1) n : The number of cells in each battery group.
- (2) m : The number of battery groups.
- (3) SOC_i : The SOC of the i th cell in the battery group.
- (4) SOC_{gj} : The average SOC of the j th battery group.
- (5) SOC_{avg} : The average SOC of all the $n \times m$ battery cells.
- (6) v_i : The output voltage of the i th cell in the battery group.
- (7) V_{gj} : The output voltage of the j th battery group (the input voltage of the j th DC/DC converter).
- (8) V_{cj} : The output voltage of the j th DC/DC converter.
- (9) V_{out} : The output voltage of the battery pack.
- (10) α_j : The duty cycle of the MOSFET in the j th DC/DC converter.
- (11) k_j : The power coefficient of the j th battery group.
- (12) ε_1 : The permissible SOC error limit in the battery group.
- (13) ε_2 : The permissible SOC error limit between the battery groups.

2. Double Layer Equalization Model

2.1. Inner Layer: Reconfigurable Equalization Topology

As shown in Figure 1, the series reconfigurable balancing group is composed of n lithium-ion battery cells and $2n$ MOSFETs. Each cell is fitted with two MOSFETs. By turning on (or off) one of the switches and turning off (or on) the other, the corresponding cell can be connected to the circuit or be isolated.

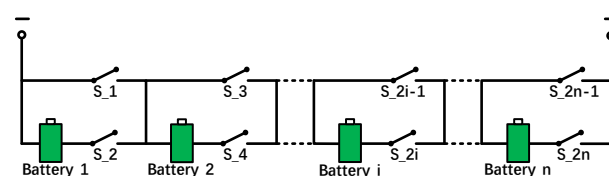


Figure 1. Reconfigurable equalization topology.

Accurate SOC estimation is critical to ensure the safe and reliable operation of battery management systems in all applications. Scholars have proposed a variety of effective approaches to estimate the SOC of the battery and diagnose faults in the circuit [25–29]. Since it is not the focus of this research, we used the measuring port of the battery cell block in Simulink to obtain the value of SOC directly.

Assume that in the initial state, SOC_1 is the highest in the group, SOC_i is the lowest in the group, and SOC_n lies between them, as shown in Figure 2. To simplify the explanation, we use cell 2 to represent the other cells in the group and assume that SOC_2 equals SOC_1 (Actually, the two SOC values do not have to be completely the same. The SOC values can be considered equal if the difference between them is less than ε_1). If each cell has a unique SOC, the system will perform this process several times until the SOC values of all cells are equal.

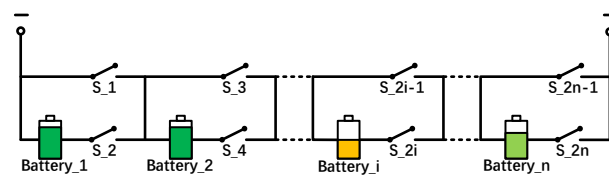


Figure 2. Initial state of SOC.

The equalization process is shown in Figure 3.

- (1) At the beginning of the equalization (Figure 3a), the system collects the SOC of all the cells in the group. The cell with the lowest SOC (cell i) is isolated (switch on S_{2i-1} , switch off S_{2i}), and the other cells are connected in series to discharge (switch on $S_2, S_4 \dots S_{2n}$, switch off $S_1, S_3 \dots S_{2n-1}$).
- (2) The series battery group continuously discharges until the SOC of the cell with the highest SOC in step (1) (cell 1) equals the SOC of the cell being isolated (cell i) (Figure 3b). Then the cell being isolated (cell i) is connected to the working part of the group (switch on S_{2i} , switch off S_{2i-1}). Meanwhile, the cell with the lowest SOC at this moment (cell n) is isolated (switch on S_{2n-1} , switch off S_{2n}).
- (3) The series battery group repeats steps (1) and (2) until all the SOC are equal. (Figure 3c). At this moment, all the cells have similar SOC, and the process of equalization in this group is over. Then all the cells are connected in series to discharge (switch on S_{2n} , switch off S_{2n-1}).

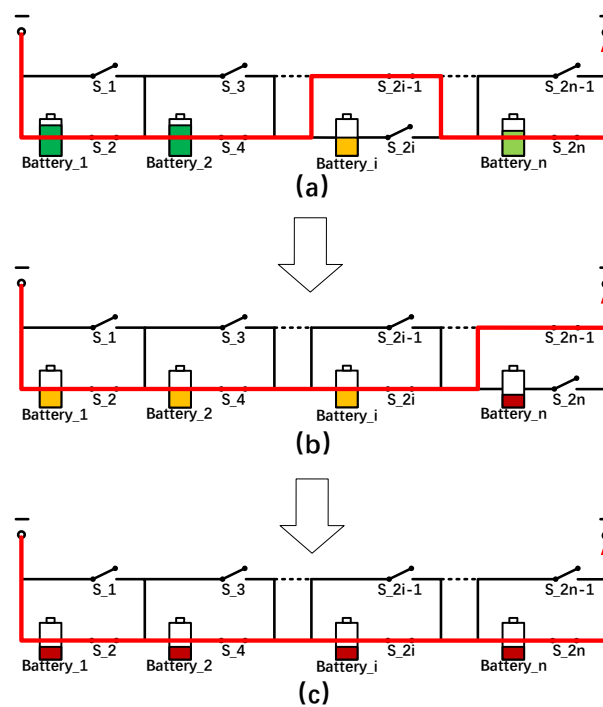


Figure 3. Equalization process: (a) At the beginning of the equalization, battery i is isolated; (b) In the middle of the equalization, battery n is isolated; (c) End of equalization.

The cell with the lowest SOC is bypassed, and the rest of the cells are connected in series to discharge during the whole equalization process. All the cells are at work state when the equalization is completed. The working part of the group consists of $n - 1$ cells during equalization while n cells after equalization. When the cells are bypassed, the output voltage of the group gets smaller, and the output current will increase at the same time to ensure constant output power. Ref. [30] has proven that the larger the output current is, the smaller the capacity the battery can discharge. In this study, no more than one cell can

be bypassed at the same time to reduce the influence of bypassing the cell on the actual discharge capacity.

Although this series reconfigurable topology can realize the equalization of the battery group, there are also some problems in practical application: firstly, the output voltage of the battery group during the equalization is the sum of $n - 1$ cell voltages.

$$V_g = \sum_i^n v_i - v_{\min} \quad (1)$$

The output voltage of the battery group after the equalization is the sum of n cell voltages.

$$V_g = \sum_i^n v_i \quad (2)$$

Therefore, there will be a voltage rise of $v_{\min}$ between before and after equalization.

Secondly, the output voltage of a lithium-ion battery decreases slowly during the discharge process [31], according to the discharge voltage curve of the lithium-ion battery (Figure 4 takes the lithium-ion battery with a rated voltage of 7.2 V, a capacity of 5.4 Ah, and a discharge rate of 0.43C as an example).

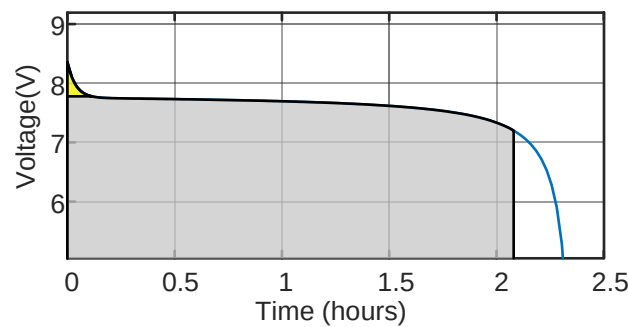


Figure 4. Discharge voltage curve.

The output voltages are 7.732 V at 0.5 h and 7.388 V at 2 h in Figure 4, and the voltage decay exceeds 4%. We can conclude that the reconfigurable topology cannot be used alone because it cannot keep a stable output voltage for the load. Therefore, it is necessary to design an outer equalization layer to achieve balance among the battery groups as well as to maintain the stability of the output voltage.

2.2. Outer Layer: Power Distribution through Buck–Boost Converters

As shown in Figure 5, the outer layer equalization topology is composed of m series-connected battery groups connected to m Buck–Boost converters, and the outputs of m converters are connected in series to supply stable power to the load.

There are m converters connected in series with the same output current, so the converter with larger output voltage V_{cj} ($j = 1, 2 \dots m$) has larger output power. According to the working principle of the Buck–Boost converter, the relationship between the output voltage and the input voltage of the j th converter is given below

$$V_{cj} = \frac{t_{\text{on}}}{t_{\text{off}}} V_{gj} = \frac{t_{\text{on}}}{T - t_{\text{on}}} V_{gj} = \frac{\alpha_j}{1 - \alpha_j} V_{gj} \quad (3)$$

where t_{on} is the on time of the MOSFET, t_{off} is the off time of the MOSFET and T is the switching period of the MOSFET. α_j can be calculated based on (4)

$$\alpha_j = \frac{V_{cj}}{V_{gj} + V_{cj}} \quad (4)$$

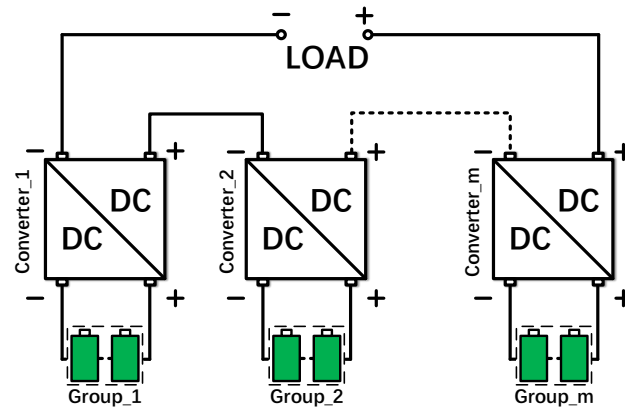


Figure 5. Topology of outer layer.

The output voltage of the converter can be controlled by adjusting the duty ratio α_j of the converter. We arrange the battery groups in descending order of SOC_{gj} ($j = 1, 2 \dots m$), which is the average SOC of the j th battery group. The converters connected to the battery groups with higher average SOC are set with higher duty ratios and output higher voltages and powers; the converters connected to the battery groups with lower average SOC are set with lower duty ratios and output lower voltages and powers. The relationship between the output voltages V_{outj} ($j = 1, 2 \dots m$) can be expressed as

$$V_{c1} \geq V_{c2} \geq \dots \geq V_{cm} \quad (5)$$

$$V_{c1} + V_{c2} + \dots + V_{cm} = V_{out} \quad (6)$$

k_j is the proportion of the output voltage of the j th converter of the total output voltage.

$$k_j = \frac{V_{cj}}{V_{out}} \quad (7)$$

The relationship between k_j ($j = 1, 2 \dots m$) can be expressed as

$$k_1 \geq k_2 \geq \dots \geq k_m \quad (8)$$

$$k_1 + k_2 + \dots + k_m = 1 \quad (9)$$

From (4) and (7), the relationship between k_j and α_j can be inferred as follows:

$$\alpha_j = \frac{k_j \cdot V_{out}}{V_{gj} + k_j \cdot V_{out}} \quad (10)$$

From (3) and (10), we can control the output voltage and power of the j th converter by adjusting α_j by calculating the appropriate k_j according to the requirement of balancing speed, V_{gj} , and V_{out} . Until the average SOC of all groups are equal (The average SOC do not have to be completely the same. The average SOC can be considered equal if the difference between any two average SOC is less than ε_2), all battery groups are set to discharge at the same power as (11).

$$k_1 = k_2 = \dots = k_m = \frac{1}{m} \quad (11)$$

The equalization of the outer layer is completed.

3. Control Strategy

3.1. Control Strategy of the Inner Layer

The conventional equalization strategy always isolates the battery cell with the lowest SOC from the group. The working part is formed by the remaining $n - 1$ cells connected in series to discharge. When the SOC of the isolated cell is equal to the cell with the highest SOC among the remaining cells, the new cell with the lowest SOC will be isolated, and the original isolated cell will be connected to the working part. The conventional equalization strategy is shown in the shaded area (dotted line) of Figure 6. This control strategy has disadvantages in that the system is always in the circulation of equalization, the cells in the working part are constantly switched, and there is always a cell isolated from the pack, which reduces both the balancing efficiency and the maximum output power of the battery group [32].

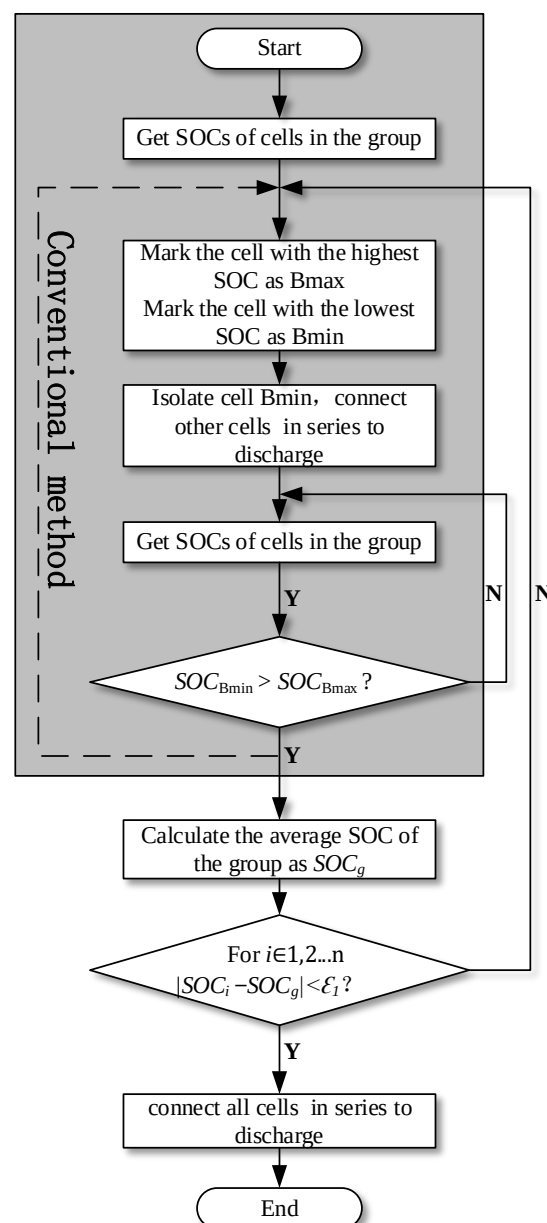


Figure 6. Control strategy of conventional method (shaded area, dotted line)/control strategy of the proposed inner layer.

In this study, an advanced control strategy is proposed. A monitoring section has been added to check whether the equalization of the inner layer is completed. It is considered that the equalization of the battery group is accomplished if the difference between the SOC of any cell in the group and the average SOC of the group is less than ε_1 . Then all n cells are connected in series to discharge.

3.2. Control Strategy of the Outer Layer

The control strategy of the outer layer proposed in this paper is shown in Figure 7. We arrange the battery groups in descending order of SOC_{gj} ($j = 1, 2 \dots m$) as mentioned in Section 2. The power coefficients k_j ($j = 1, 2 \dots m$) are set in descending order as well. SOC_{avg} is the average SOC of all the $n \times m$ cells. It is considered that the j th battery group has reached a balance if the difference between SOC_{gj} and SOC_{avg} is less than ε_2 after a period of equalization. After all m battery groups get balanced, the same power coefficient of $1/m$ is set for all battery groups, and all battery groups discharge at the same power. The equalization of the system is completed.

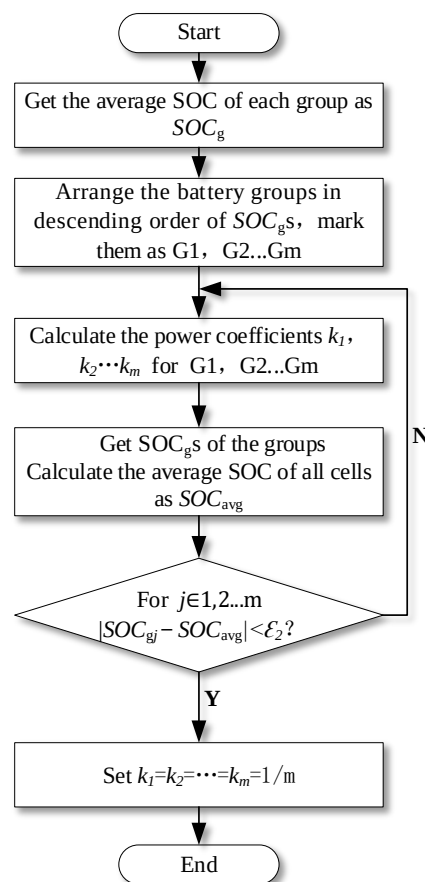


Figure 7. Control strategy of the proposed outer layer.

4. Calculation and Comparison

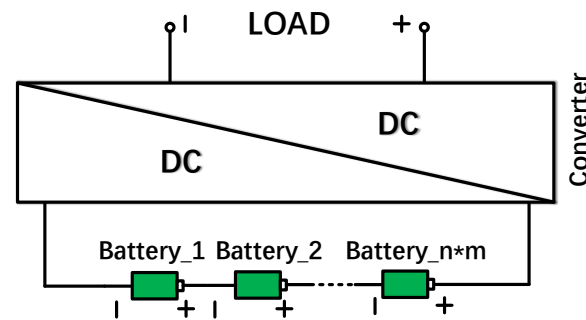
In this section, calculations are made to compare the efficiency of the proposed method with the conventional method. Some of the parameters used in the calculations are listed in Table 1 (ignoring the resistance of the lead).

Table 1. The parameters of equalization efficiency calculations.

Parameters	Value
number of battery cells $n \times m$	4×3
internal resistance of a battery cell r/Ω	0.01
internal resistance of a MOSFET r_s/Ω	0.001
output voltage of a battery cell v/V	7.2
load voltage V_{out}/V	72
load current I_{out}/A	1
load power P/W	72
equalization power P_{eq}/W	
efficiency of DC/DC converter $\eta/\%$	

4.1. Efficiency Calculation of the Ideal Working Circuit

Firstly, an ideal series working circuit will be introduced, as shown in Figure 8. It is assumed that the battery cells in the ideal circuit are balanced all the time, and only the power loss caused by the internal resistances of the cells and the efficiency of the converter should be taken into consideration.

**Figure 8.** Ideal circuit.

P_g denotes the output power of the series battery group (the input power of the converter). The relationship between P_g and P is

$$P_g = \frac{P}{\eta} \quad (12)$$

The output voltage of the battery group can be calculated as

$$V_g = nmv \quad (13)$$

The current flowing through each cell can be calculated as

$$I_g = \frac{P_g}{V_g} = \frac{P}{nmv\eta} \quad (14)$$

The power loss caused by the internal resistances of the cells is

$$P_r = I_g^2 R = \left(\frac{P}{nmv\eta}\right)^2 nmr = \frac{P^2 r}{nmv^2 \eta^2} \quad (15)$$

The total power all the battery cells spend is

$$P_{all} = P_g + P_r \quad (16)$$

Thus, the total efficiency of the ideal circuit can be expressed as

$$\eta_1 = \frac{P}{P_{all}} = \frac{1}{\frac{1}{\eta} + \frac{Pr}{nmv^2 \eta^2}} \quad (17)$$

4.2. Efficiency Calculation of the Conventional Equalization Circuit Based on DC/DC Converter

A conventional active equalization circuit using a converter as an energy transfer device is shown in Figure 9. This circuit consists of two DC/DC converters and $n \times m$ battery cells connected in series. DC/DC converter 1 equalizes the cells in the pack, while DC/DC converter 2 controls the output voltage to be stable.

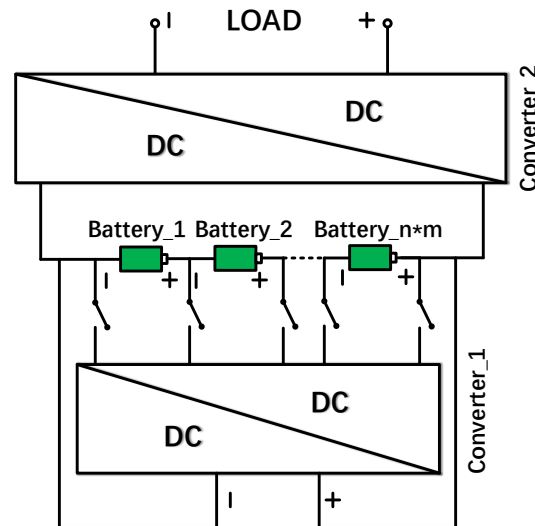


Figure 9. Conventional converter equalization circuit.

The output power, the output voltage of the series battery group, and the current flowing through the cells of the conventional converter equalization circuit can also be expressed as Equations (12)–(14). The power loss caused by the internal resistances of the cells and the MOSFETs of the conventional converter equalization circuit can be calculated as

$$P_{eqr} = I_{eq}^2 \cdot 2r_s = \left(\frac{P_{eq}}{v}\right)^2 2r_s \quad (18)$$

Moreover, the power loss of converter 1 during the equalization is

$$P_{cvt} = P_{eq}(1 - \eta) \quad (19)$$

The total power all the battery cells spend is

$$P_{all} = P_g + P_r + P_{eqr} + P_{cvt} \quad (20)$$

The total efficiency of the conventional converter equalization circuit can be expressed as

$$\eta_2 = \frac{P}{P_{all}} = \frac{P}{\frac{P}{\eta} + \left(\frac{P}{nmv\eta}\right)^2(nmr + 2r_s) + P_{eq}(1 - \eta)} \quad (21)$$

4.3. Efficiency Calculation of the Proposed Circuit

The total output power of the series battery groups of the proposed circuit can also be expressed as (12). The output voltage of a single battery group in the proposed circuit during the equalization is

$$V_g = (n - 1)v \quad (22)$$

The current flowing through each cell can be calculated as

$$I_g = \frac{P_g}{mV_g} = \frac{P}{(n - 1)mv\eta} \quad (23)$$

The power loss caused by the internal resistances of the cells and the MOSFETs of the proposed circuit can be calculated as

$$P_r = I_g^2 R = \frac{P^2[(n-1)r + nr_s]}{(n-1)^2 m v^2 \eta^2} \quad (24)$$

The total power all the battery cells spend is

$$P_{\text{all}} = P_g + P_r \quad (25)$$

The total efficiency of the proposed equalization circuit can be expressed as

$$\eta_3 = \frac{P}{P_{\text{all}}} = \frac{1}{\frac{1}{\eta} + \frac{P[(n-1)r + nr_s]}{(n-1)^2 m v^2 \eta^2}} \quad (26)$$

4.4. Comparison of Balancing Efficiency

The efficiencies of the three circuits mentioned above are compared under different P_{eq} and η .

Firstly, assuming that η is constant 85%, the efficiencies of the three circuits versus P_{eq} are shown in Figure 10. The efficiency of the conventional converter equalization circuit is approximate to that of the ideal circuit and higher than the efficiency of the proposed circuit when the equalization power P_{eq} is close to zero. In this situation, the proposed circuit performs worse than the conventional converter circuit because of the extra energy loss caused by MOSFETs. However, the efficiency of the conventional converter circuit decreases straightly with the increase of P_{eq} , while the other two circuits maintain the same efficiency. The proposed circuit outperforms the conventional converter circuit when P_{eq} is larger than 0.5 W in this case. Compared to the conventional converter circuit, the proposed circuit improves the efficiency more significantly when the imbalance gets bigger.

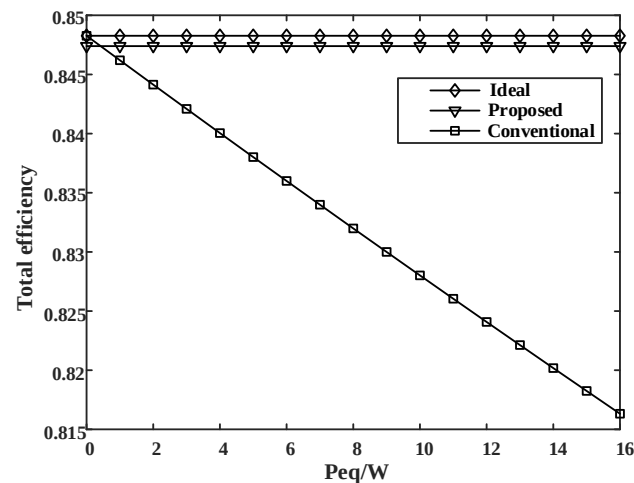


Figure 10. The total efficiency versus the equalization power P_{eq} .

Assuming that P_{eq} is constant 10 W, the efficiencies of the three circuits versus η are shown in Figure 11. The efficiencies of the three circuits are close to 100% when η is close to 100%. The total efficiency of the three circuits all decrease with the decrease of η . The efficiency of the proposed circuit is always close to that of the ideal circuit, but the efficiency of the conventional converter circuit decreases faster than those of the other two circuits. Compared to the conventional converter circuit, the smaller η is, the greater the efficiency improvement the proposed circuit makes.

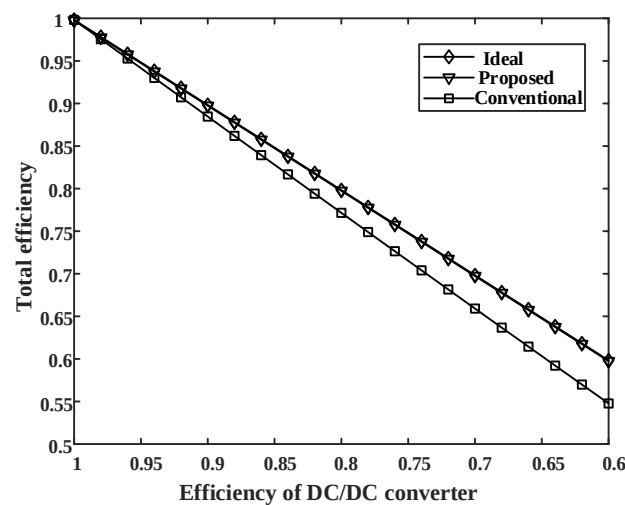


Figure 11. The total efficiency versus the efficiency of converter η .

5. Simulation Results and Discussion

The simulation of a 12 battery cell circuit built on the Matlab/Simulink platform was carried out to validate the feasibility of the proposed double-layer reconfigurable equalization method, as shown in Figure 12. There were three battery groups connected to three Buck–Boost converters, respectively, to form the outer layer. In the inner layer, each battery group consisted of four lithium-ion battery cells connected in series. The three converters in the outer layer were connected in series to supply power to the load which was set to be 50 Ω .

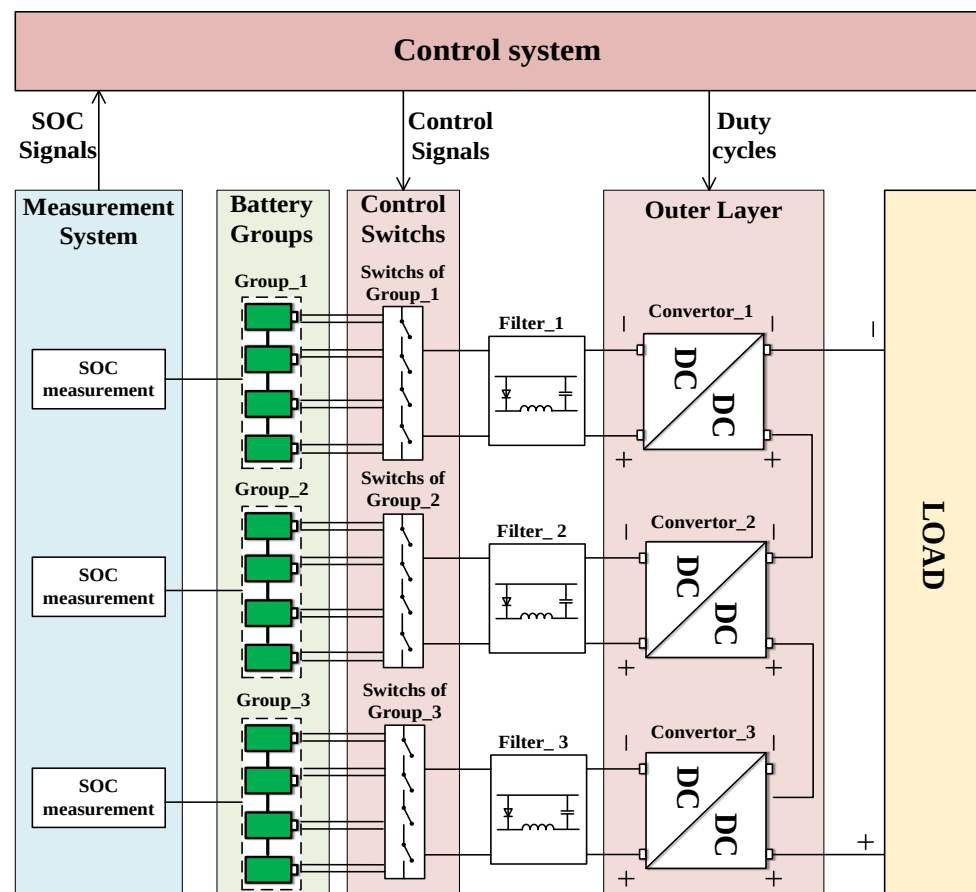


Figure 12. Schematic diagram of proposed method.

Each lithium-ion battery cell was set to have a rated voltage of 7.2 V and a rated capacity of 5.4 Ah. In the initial state, the SOC of cell 1 to cell 12 are shown in Table 2. The expected load voltage was set to 150 V.

Table 2. The SOC of battery cells.

Cell Number	1	2	3	4	5	6	7	8	9	10	11	12
SOC (%)	100	99.96	99.92	99.88	99.84	99.80	99.76	99.72	99.68	99.64	99.60	99.56

5.1. Simulation Results of the Inner Layer

The permissible SOC error limit in the battery group, ε_1 , was set to 0.001%. Battery group 1 consisted of cell 1, cell 2, cell 3, and cell 4. The initial SOC of the cells were 100%, 99.96%, 99.92%, and 99.88%. The equalization process of the inner layer is shown in Figure 13.

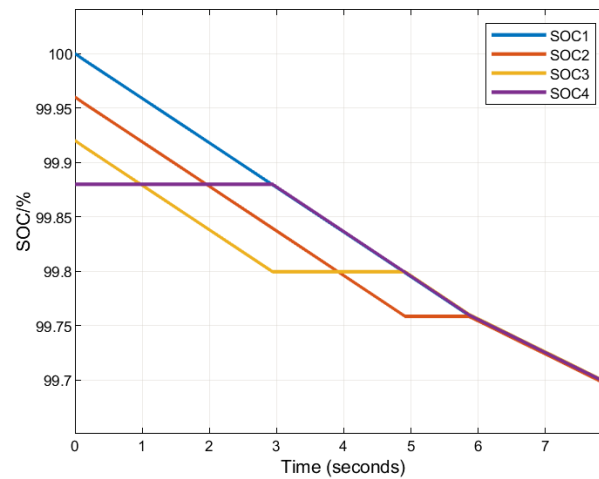


Figure 13. Equalization process of battery group 1.

Cell 1, cell 2, and cell 3 had higher SOC and discharged first, while cell 4, with the lowest SOC, was isolated at the beginning of equalization. Cell 4 got balanced and was set to discharge at 2.92 s (the time the difference between SOC_4 and SOC_1 is no more than ε_1); at the same time, cell 3 with the lowest SOC got isolated. The equalization processes of cell 3 and cell 2 were similar to that of cell 4. Cell 3 and cell 2 reached balance at 4.90 s and 5.87 s. The inner layer equalization of battery group 1 was completed, and all the cells were connected in series to discharge at 5.87 s. There were three cells discharging in the group before 5.87 s but four cells after 5.87 s. The output power of each cell was reduced because the output power of the battery group was constant, but the number of working cells increased. Therefore, the decline rate of the SOC curve slowed down after 5.87 s.

5.2. Simulation Results of the Outer Layer

The permissible SOC error limit between the battery groups, ε_2 , was set to 0.002%. The average SOC of battery groups 1, 2, and 3 were 99.94%, 99.78%, and 99.62% in the initial state. In this case, we have three battery groups, so we should set k_1 , k_2 , and k_3 according to Equation (27) to make sure the three battery groups get balanced at the same time.

$$\frac{k_1 - k_2}{k_2 - k_3} = \frac{SOC_{g1} - SOC_{g2}}{SOC_{g2} - SOC_{g3}} \quad (27)$$

The three battery groups discharge at different powers according to k_1 , k_2 , and k_3 . The equalization speed of the outer layer depends on the ratio of k_1 and k_3 . The equalization process of the outer layer is shown in Figure 14.

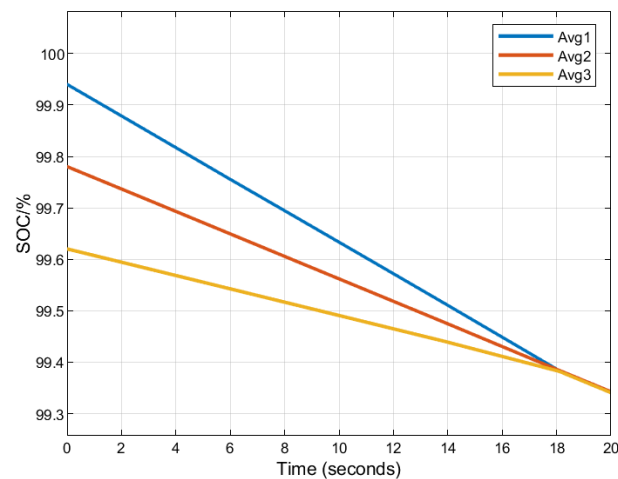


Figure 14. Equalization process of outer layer.

The power coefficients k_1 , k_2 , and k_3 were calculated and set to be 7/15, 5/15, and 3/15. The differences between any two average SOC of the three battery groups reached less than 0.002% at 18.1 s. At that moment, the equalization of the outer layer was finished. k_1 , k_2 , and k_3 were reset to 1/3 so that all battery groups would discharge at the same power.

5.3. Simulation Results of the Battery Pack

Each battery group in the outer layer outputs different power due to the different power coefficients k . When the imbalance degrees of the groups are the same, which means the groups have the same amount of electricity to balance, the higher the output power is, the faster the battery group accomplishes its equalization. The equalization process of the battery pack is shown in Figure 15.

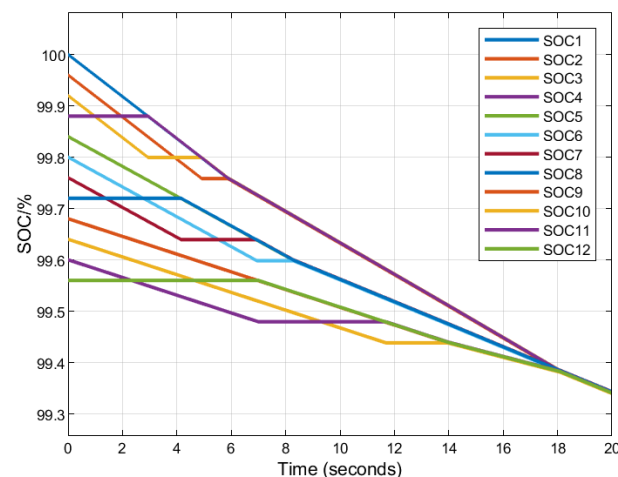


Figure 15. Equalization process of battery pack.

Battery group 1, with the highest output power, got balanced at 5.87 s first. Battery group 2, with lower output power, got balanced at 8.28 s. Battery group 3, with the lowest output power, got balanced at 13.96 s. All battery groups in the inner layer were balanced, and all cells were connected to the circuit at 13.96 s. The system finished its equalization, and all the groups began to discharge at the same power at 18.1 s.

5.4. Simulation Results of the Output Voltage

Each battery group was connected to a Buck–Boost converter, and the outputs of the converters were connected in series. The output voltages of the converters are shown

in Figure 16. The output voltages of the three converters became stable after a rise of about 0.2 s. The calculated steady-state voltages of converters 1, 2, and 3 are 70 V, 50 V, and 30 V according to Equation (7), and the simulation results are consistent with the calculation results.

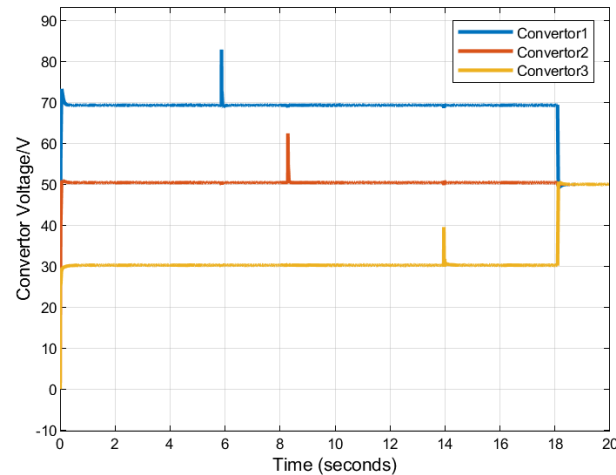


Figure 16. Output voltage of each converter.

The total output voltage of the battery pack was generally stable at 150 V, as shown in Figure 17. The output voltage had pulse distortion at 5.87 s, 8.28 s, and 13.96 s (all of which returned stabilization within 0.1 s), as shown in Figures 16 and 17. The reason is that the number of working cells in the group increased from three to four when the equalization in the battery group was completed. The input voltage of the converter instantly increased by 1 cell voltage. However, the input current and the output current of the converter could not suddenly change due to the inductance in the converter. The output power of the converter would rise briefly and then stabilize. The number of battery groups will be far greater than three, and the number of battery cells in each group will be far greater than four when the proposed method is used in practical energy storage systems. So the voltage distortion caused by the switch of one battery cell will be small enough to be ignored. There are also some ways to eliminate the distortion. We can eliminate the voltage distortion and ensure the stable output voltage of the system by connecting the extra cells to the groups during equalization and isolating them after equalization. The same result can be achieved by using a regulated power supply.

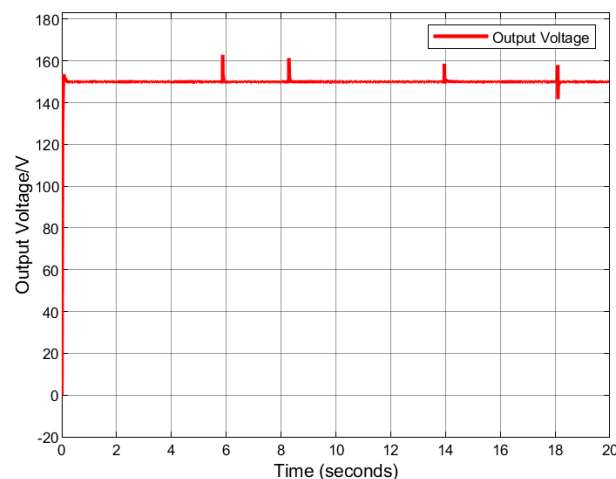


Figure 17. Output voltage of the battery pack.

6. Conclusions

In this paper, a double-layer equalization method combining the reconfigurable topology with the converter active equalization method is proposed. The equalization strategies of the inner and outer layers are discussed in detail, and the feasibility of the proposed method is validated by simulation. The main concluding remarks can be made below:

- (1) The inner layer proposed in this paper realizes balance among cells in the same group through the reconfigurable topology. Compared with the conventional active equalization method, the proposed method uses fewer components, has lower cost, higher efficiency, and simpler control complexity. Moreover, the equalization process goes on equally well regardless of the positions the cells needed to be balanced in the group.
- (2) The method of connecting the outputs of converters in series is used in the outer layer. The proposed outer layer can realize the balance among battery groups and keep the output voltage and power of the system stable at the same time. It solves the problem that using the reconfigurable topology alone cannot stabilize or adjust the output voltage.
- (3) The efficiency of the proposed equalization method was calculated and compared with the conventional converter active equalization method. The results showed that the proposed method can improve the balancing efficiency, and the higher the unbalance of the battery system is, the more significant the improvement is. The proposed method is suitable for scenarios like retired battery energy storage systems which have high power, large capacity, and high requirements for balancing efficiency and robustness.

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